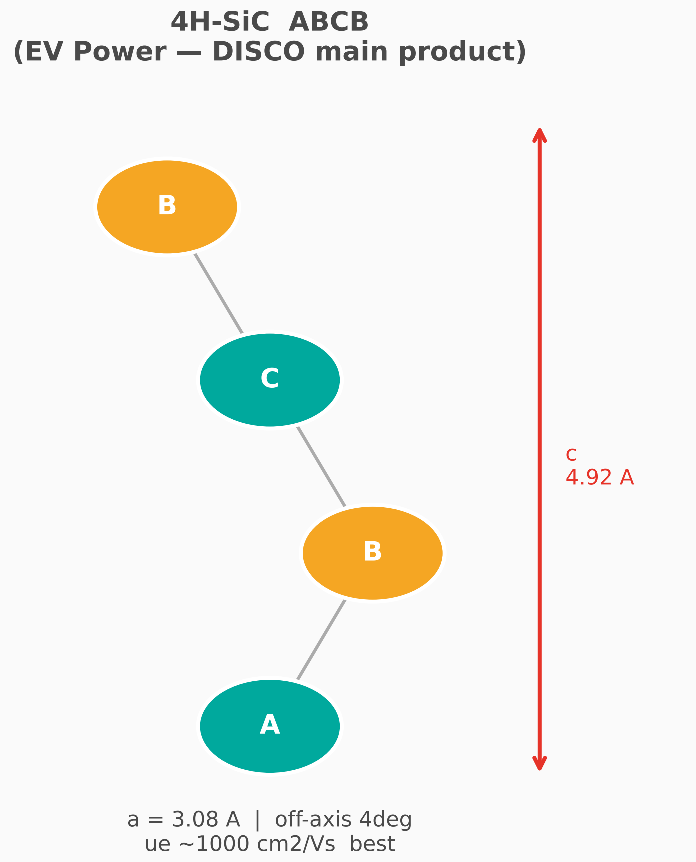
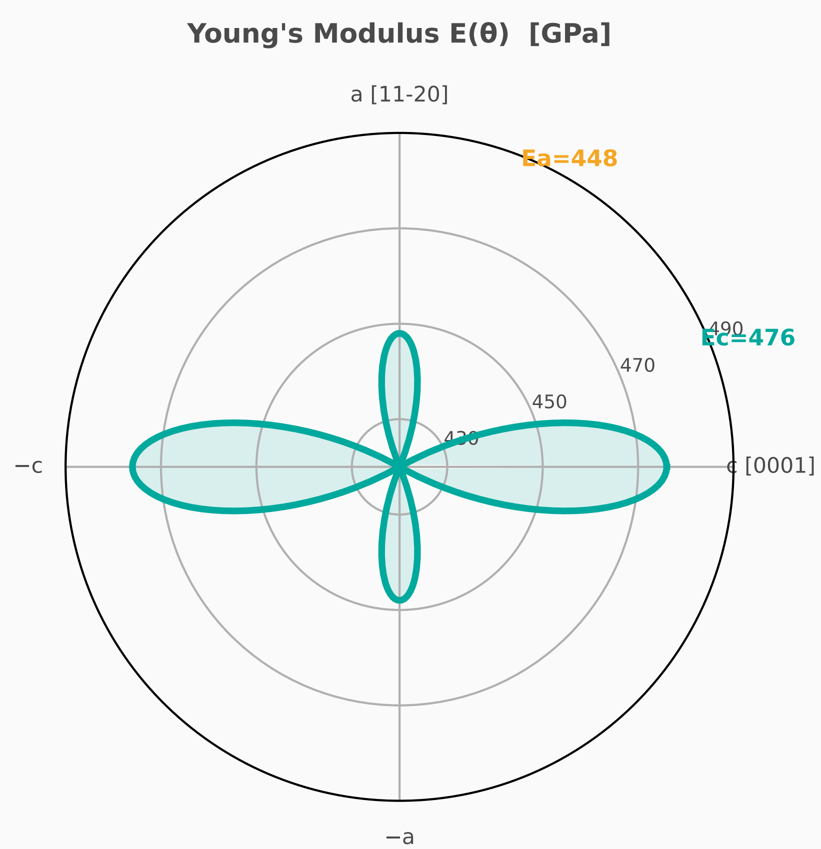
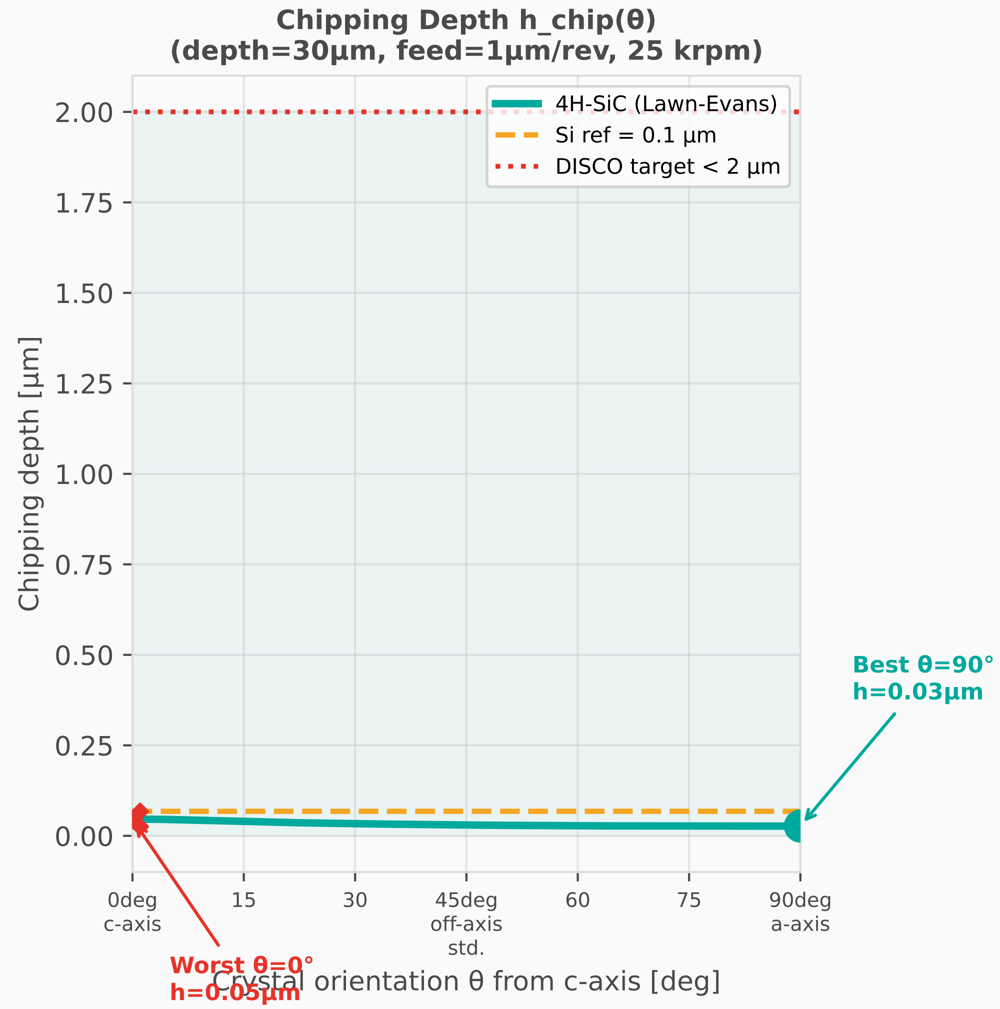
4H-SiC Crystal Anisotropy · Dicing Chipping Model







4H vs 6H vs Si: Key Properties

Property	4H-SiC	6H-SiC	Si
Bandgap [eV]	**3.26**	3.03	1.12
ue [cm2/Vs]	**~1000**	~400	1400
u isotropy	**Good**	Poor x100	Isotropic
Ec [GPa]	**476**	~501	130
Ea [GPa]	**448**	~448	130
K_IC [MPa*sqrtm	**1.0 - 2.8**	1.0 - 2.8	0.70
Hardness [GPa]	**26 - 28**	26 - 28	13
Etchrate (KOH)	**Resistant**	Resistant	Fast
EV power devic	**Best (main)**	Not for EV	--
GaN-on-X sub	**Good (growing**	Legacy	No
DISCO cut vo	**Rapidly growin**	Almost none	High
Blade wear rate	**2x Si**	2x Si	Baseline